

where $v = 0$, the spaceship will be travelling toward the planet.

- 11 A satellite of mass 200 kg is moved from an orbit where the gravitational potential is -80 MJ kg^{-1} to another orbit where the gravitational potential is -40 MJ kg^{-1} .

What is the increase in its **total** energy?

- A 0 MJ B 2000 MJ C 4000 MJ D 8000 MJ

Ans: C

$$U = -\frac{GMm}{r}$$

$$TE = -\frac{GMm}{2r} = \frac{U}{2} = \frac{m\phi}{2}$$

$$\Delta TE = 0.5m(\phi_f - \phi_i) = 0.5(200)(-40 - (-80)) = 4000 \text{ MJ}$$

12 A small ice cube of mass 20 g is heated and changes from the solid to the liquid state. During